3. Write a C program to find ε -closure for all the states in a Non-Deterministic
Finite Automata (NFA) with ε -moves

Aim

To write a C program to find the ε-closure of all states in a Non-Deterministic Finite Automaton (NFA) with ε-moves.

Procedure

1. Start the program.
2. Read the number of states and the number of ε-transitions.
3. Store the ε-transitions in an adjacency matrix.
4. For each state, initialize its ε-closure with the state itself.
5. Follow all possible ε-transitions from that state and add the reachable states to its ε-closure.
6. Repeat until no new states can be added.
7. Display the ε-closure of every state.
8. Stop the program.

Program:

```
#include <stdio.h>

int main() {
    int n, e, i, j, u, v, k;
    int a[20][20] = {0};
    int closure[20][20] = {0};

    printf("Enter number of states: ");
    scanf("%d", &n);

    printf("Enter number of epsilon transitions: ");
    scanf("%d", &e);

    printf("Enter transitions (from to):\n");

    for (i = 0; i < e; i++) {
        scanf("%d %d", &u, &v);
```

```
        a[u][v] = 1;
    }

    for (i = 0; i < n; i++) {
        closure[i][i] = 1;

        for (j = 0; j < n; j++) {
            if (a[i][j])
                closure[i][j] = 1;
        }
    }

    for (k = 0; k < n; k++) {
        for (i = 0; i < n; i++) {
            for (j = 0; j < n; j++) {
                if (closure[i][k] && closure[k][j])
                    closure[i][j] = 1;
            }
        }
    }

    printf("\nEpsilon-closure of each state:\n");

    for (i = 0; i < n; i++) {
        printf("E-closure(q%d) = { ", i);

        for (j = 0; j < n; j++) {
            if (closure[i][j])
                printf("q%d ", j);
        }

        printf("}\n");
    }

    return 0;
}
```

Result

Thus, the C program to find the ε-closure of all states in an NFA with ε-moves was written and executed successfully.